

Quasicrystalline Emergence v7.3 (Expanded)

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October 27, 2025

Abstract

We update the *Quasicrystalline Emergence* framework with code-backed, reproducible methods: (i) a corrected mapping between the frustration parameter $\delta\delta$ and the phantom-crossing redshift z_{cross} ; (ii) a small- $\delta\delta$ analytic expression for $H(z)$; (iii) a minimal EFT stability region for a k-essence realization; (iv) log-periodic templates for spectra induced by the golden-ratio clock; (v) calibrated gravity-induced decoherence scales; and (vi) an explicit $E8 \rightarrow 3DE8 \rightarrow 3D$ cut-and-project construction with an acceptance-window *autotuning* rule that yields clear icosahedral signatures. All items are falsifiable and shipped with a one-line CLI to reproduce tables and plots.

Version note (7.3): Adds autotuning/CLI, consolidates the cut-and-project rule, and includes reproducibility statements and figure stubs.

Setup and Notation

We use $(\Omega_m, \Omega_\Lambda) = (0.3, 0.7)$ when numbers are shown, with H_0 left explicit. The golden ratio is $\varphi = 1 + \sqrt{5}/2$ and $\ln \varphi \approx 0.4812118251$. Matter- Λ equality occurs at

$$a_{\text{eq}} = (\Omega_m \Omega_\Lambda)^{1/3} \approx 0.753947441, z_{\text{eq}} = a_{\text{eq}}^{-1} \approx 0.32605. \quad a_{\text{eq}} = (\Omega_\Lambda \Omega_m)^{1/3} \approx 0.753947441, z_{\text{eq}} = a_{\text{eq}}^{-1} \approx 0.32605.$$

Define the **golden-ratio time**

coordinate

$$\theta(a) \equiv \ln(a/a_{\text{eq}}) / \ln \varphi. \quad \theta(a) \equiv \ln \varphi \ln(a/a_{\text{eq}}).$$

The dark-energy density is expanded near Λ
as Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

$$\rho_{de}(a) = \rho_{\Lambda} [1 + \varepsilon \theta + \nu \theta^2 + O(\theta^3)], \delta \equiv \varepsilon 2\nu, p_{de}(a) = p_{\Lambda} [1 + \varepsilon \theta + \nu \theta^2 + O(\theta^3)], \delta \equiv 2\nu \varepsilon,$$

with small curvature parameters (ε, ν) ; the case $\varepsilon = \nu = 0$ reduces to Λ CDM. Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

Corrected Phantom-Crossing Redshift

The equation of state near Λ
is Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

$$w(a) \approx -1 - \varepsilon + 2\nu \theta(a) 3 \ln \varphi. w(a) \approx -1 - 3 \ln \varphi \varepsilon + 2\nu \theta(a).$$

The crossing $w = -1$ happens at $\varepsilon + 2\nu \theta = 0$, i.e. at $\theta_{cross} = -\delta \theta_{cross} = -\delta$,
whence Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

$$z_{cross} = a_{cross}^{-1} - 1 = a_{\Lambda eq}^{-1} \varphi^{\delta} - 1. z_{cross} = a_{cross}^{-1} - 1 = a_{\Lambda eq}^{-1} \varphi^{\delta} - 1.$$

Sign fix: This equation implies that hitting $z_{cross} < z_{\Lambda eq}$ requires *negative* δ . For the numerical target $z_{cross} = 0.29$ we obtain Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

$$\delta = \ln[(1 + z_{cross}) a_{\Lambda eq}] \ln \varphi \approx -0.0577508704. \delta = \ln \varphi \ln[(1 + z_{cross}) a_{\Lambda eq}] \approx -0.0577508704.$$

Small- δ Analytic Background

Write $u \equiv \ln(1+z)$ and use $\ln a = -u$. From the dark-energy density expansion and equation of state, the dark-energy density relative to ρ_{Λ} integrates to Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

$$\ln \rho_{de}(z) \rho_{\Lambda}^{-1} \approx -\varepsilon \ln \varphi + \nu (\ln \varphi)^2 [u^2 + 2(\ln a_{\Lambda eq}) u] + O(\delta^3) \ln \rho_{\Lambda} p_{de}(z) \approx -\ln \varphi \varepsilon u + (\ln \varphi)^2 \nu [u^2 + 2(\ln a_{\Lambda eq}) u] + O(\delta^3)$$

and the background
becomes Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt

$$H^2(z) H_0^2 = \Omega_m (1+z)^3 + \Omega_{\Lambda} \exp[-\varepsilon \ln \varphi + \nu (\ln \varphi)^2 (u^2 + 2(\ln a_{\Lambda eq}) u)] + O(\delta^3). H_0^2 H^2(z) = \Omega_m (1+z)^3 + \Omega_{\Lambda} \exp[-\ln \varphi \varepsilon u + (\ln \varphi)^2 \nu (u^2 + 2(\ln a_{\Lambda eq}) u)] + O(\delta^3).$$

This closed form gives distances by a single quadrature. With $(v, \delta) = (0.10, -0.0577509)$ we have $\varepsilon = 2v\delta \simeq -0.0115502$ and the $w = -1$ crossing lands at $z \simeq 0.29$.

Minimal EFT Realization and Stability

A one-field k-essence near the background $X_0 = -12(\partial\Phi)^2$ with

$$K(X) \approx X + \alpha (X - X_0)^2$$

satisfies, at X_0 ,

$$KX = 1, KXX = 2\alpha, c_s^2 = KXX + 2X_0KXX = 11 + 4\alpha X_0.$$

The **stability region** is therefore

$$\alpha X_0 > -1/4 \text{ (no ghost/gradient)}, 0 < c_s^2 = 11 + 4\alpha X_0 \leq 1 \Leftrightarrow \alpha X_0 \geq 0.$$

A two-field completion can realize a continuous $w = -1$ crossing while keeping each sector ghost-free.

Log-Periodic Templates from the φ -Clock

Discreteness in $\theta = \ln a / \ln \varphi$ produces log-periodic modulations. A data-ready multiplicative template for a baseline spectrum $P_0(k)$ is

$$P(k) = P_0(k) [1 + A \cos(\omega \ln(k/k_*) + \phi_0)], \omega = 2\pi \ln \varphi \approx 13.0570, \pi \approx 3.14159.$$

Amplitudes A and phase ϕ_0 are exposed by CLI/env toggles (`E8_LP_*`).

$E8 \rightarrow E8 \rightarrow$ 3D Quasicrystal Projection, Window Tuning, and Autotune

Implementation

We build the 240 $E8E8$ roots in $R8R8$, construct a $\varphi\varphi$ -loaded $8 \times 38 \times 3$ seed and obtain orthonormal projectors $(P_{\parallel}, P_{\perp})(P_{\parallel}, P_{\perp})$ via QR with an 8D completion. After cut-and-project, we set the neighbor cutoff in $X\|X\|$ to the *median distance* to the 12th nearest neighbor, which targets icosahedral coordination without forcing identical degree. `Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt`

Autotuning Rule

For a fixed projector we sweep the acceptance fraction `keep_frac` $\in \{0.25, 0.30, 0.35, 0.40, 0.45, 0.50, 0.55\}$, retaining the roots with the smallest $\|X_{\perp}\|$ and building the neighbor graph with the median-12th cutoff. For each sweep point we compute: (i) d^{-} (mean degree); (ii) the degree-12 band share f^{11-13} ; (iii) an **icosahedral angle score** `Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt`

$$A_{ico} \equiv 12 \left(\frac{|\theta - 63.4349^\circ| \leq 7.5^\circ}{\pi} + \frac{|\theta - 116.5651^\circ| \leq 7.5^\circ}{\pi} \right). A_{ico} \equiv 21 \left(\frac{|\theta - 63.4349^\circ| \leq 7.5^\circ}{\pi} + \frac{|\theta - 116.5651^\circ| \leq 7.5^\circ}{\pi} \right).$$

We select the window
`byQuasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt`

$$S = 0.70 A_{ico}^{\wedge} + 0.25 f^{11-13} - 0.05 d^{-}, S = 0.70 A_{ico} + 0.25 f^{11-13} - 0.05 d^{-},$$

where $\cdot^{\wedge}$ denotes min-max normalization across the sweep. In our construction the score peaks at `keep_frac` ≈ 0.30 . `Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt`

Reproducibility and CLI

All results are reproducible with the accompanying code. A one-line sweep: `Quasicrystalline_Emergence_v7_3_EXPANDED_texsource_extracted.txt`

text

```
python e8_mind_server_M24.5_patched_cli.py --qc-tune
```

This prints a table over `keep_frac` with $A_{\text{ico}}A_{\text{ico}}$, $f_{11-13}f_{11-13}$, d^{d} , and the chosen cutoff radius, and caches the selected window for subsequent runs. `Quasicrystalline_Emergence_v7_3_EXPANDED_tsource_extracted.txt`

Falsifiable Prediction

With the φ -seeded projector and the median-12th-neighbor cutoff, the $E8 \rightarrow 3DE8 \rightarrow 3D$ quasicrystal patch exhibits:

1. A degree distribution centered near 12
2. Enhanced neighbor-pair angles near 63.43° and 116.57°

Either feature failing to appear under the stated procedure falsifies this construction. `Quasicrystalline_Emergence_v7_3_EXPANDED_tsource_extracted.txt`

Working Theory

The results above are model-level predictions or diagnostics contingent on the quasicrystalline-emergence assumptions; each item is designed so that it can be directly falsified by data or simulation. `Quasicrystalline_Emergence_v7_3_EXPANDED_tsource_extracted.txt`

1. https://ppl-ai-file-upload.s3.amazonaws.com/web/direct-files/attachments/104752356/481ef818-adad-4cf7-a4c0-f54ff52f4910/Quasicrystalline_Emergence_v7_3_EXPANDED_tsource_extracted.txt